

HEALTHCONNECT CLINIC

PROJECT SUMMARY

1. Week 5 Plan

The Week 5 plan was to move from the Week 4 foundation into practical analysis. The intended work included data preparation and quality checks, calculation of selected KPIs, exploratory analysis of attendance and no-show patterns, segmentation across relevant variables, development of an analytical dashboard, and translation of findings into business insights and recommendations.

2. What Was Completed

The HealthConnect appointment dataset was analysed using Microsoft Power BI.

- The selected KPI framework was implemented, including No-Show Rate, Attendance Rate, Cancellation Rate, Reminder-Associated Attendance Rate and Repeat No-Show Rate.
- An interactive exploratory dashboard was developed with KPI cards and visual comparisons across previous no-shows, booking lead time, appointment type, appointment day, age group, reminder channel, distance to clinic, waiting time and appointment outcomes.

3. Key Findings

- The dashboard shows a 48.5% no-show rate, 46.3% attendance rate and 5.3% cancellation rate across 5,000 appointments.
- No-show rates increase substantially with previous no-show history and with longer booking lead times.
- Reminder-channel attendance also differs across groups, with SMS showing the highest displayed attendance rate.
- The 60+ minute waiting-time group shows a notably high no-show rate and requires further investigation.

4. Major Challenges Encountered

A key implementation challenge was creating and using custom grouping fields in Power BI, particularly for booking lead days, distance to clinic and waiting time. Another challenge was handling measure and grouping dependencies correctly so that visuals could be built without circular-dependency errors. These issues were resolved by creating explicit grouped fields/measures and using them consistently in the visuals.

5. Important Decisions Made and Why

The Week 5 analysis retained the KPI framework proposed in Week 4 because the selected measures directly address the core business problem. The analysis also used grouped variables for continuous measures such as booking lead days, distance and waiting time because grouped comparisons are easier to interpret in an exploratory business dashboard. The original dataset was treated as the source and should remain preserved separately from transformed analytical data.

6. Changes to Week 4 Approach

Week 4 proposed the analysis sequence of Prepare → Profile → Measure → Compare → Investigate → Interpret → Communicate. Week 5 implemented this approach in practice by moving from data quality understanding to KPI calculation, segmented comparisons and dashboard communication. The approach was therefore expanded rather than fundamentally changed.

7. Cross-Track Collaboration Point

The principal Data Analytics dependency is the potential use of KPI definitions and no-show patterns by the Data Science track for later predictive analysis. The Week 5 dashboard provides candidate analytical features and risk segments involving previous no-show history, booking lead time, reminders, waiting time, distance and appointment characteristics. This was relevant because these findings can help inform the selection and investigation of potential variables for future no-show prediction modelling. The collaboration will improve alignment between exploratory analysis and future modelling, while recognizing that the observed relationships require further validation.

8. Remaining Work

Further work remains to validate the strongest patterns using group counts and deeper segmentation, assess statistical strength where appropriate, refine KPI definitions, review dashboard usability, and prepare the evidence needed for subsequent predictive or operational work. Reminder effectiveness should also be tested more rigorously before making causal claims.

9. Proposed Focus for Week 6

Week 6 should focus on validating and extending the findings of Week 5. Priority areas should include deeper investigation of repeat no-shows, long booking lead times, reminder-channel differences and long waiting times; stronger validation of segment sizes; refinement of dashboard storytelling; and preparation of clean analytical outputs that can support later predictive modelling and operational decisions.